SAMPLE

Earth's Sky: Meteors and Meteor Showers

Shooting Stars

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Introduction

This volume collects the sky's most democratic spectacle: the meteor, the streak of light that costs nothing to see, needs no telescope, and turns any clear dark night into a show. The subject looks simple, a bit of dust burning in the air, and the chapters will show it is anything but: the streak is a physics lesson in plasma and ablation, the shower is a map of a comet's exhalations laid along decades of orbit, the storm is a rare alignment whose prediction required the mathematics of dust trails, and the meteorite that survives to the ground is the cheapest spacecraft mission in existence, delivering solar-system-old material to doorsteps and car trunks free of charge. The book treats the whole chain: the phenomenon's discovery history from the Chinese annals to the Chelyabinsk dashcams, the characteristics of the streams and the storms, the science of ablation and airbursts, the observer's craft from lawn-chair counts to radar and camera networks, the great showers' individual biographies, the research frontier of parent bodies and impact hazard, and the trivia shelf the subject richly deserves.

A reading note belongs at the front: this volume's sources are the sky itself and the worldwide community that watches it. The International Meteor Organization's catalogs and the photographic and radar surveys supply the numbers; the named showers' parent bodies come from comet-orbit detective work the later chapters describe; and the historical record, the Chinese annals and the European chronicles and the ship logs, supplies the storm history. The book carries results rather than citation lists, the reader's trust being earned in the chapters' arithmetic, everything here being checkable against the annual show the sky stages for free.

The volume's structure follows the series' canonical sixteen-part architecture, the chapters being the same doors the sibling books use: the discovery history, the characteristics, the science, the observation craft, the features, the contexts, the research, the discoveries, the mysteries, the future, the trivia shelf, and the significance verdict, the last being the reader's own inheritance and the first being the subject's own long biography; the reader who has walked these doors before knowing the map, the reader who is new being welcome, and the subject being the series' best gate in either case.

Chapter 1: The Discovery of Meteors and Meteor Showers

The discovery of the meteor is the story of a phenomenon that was seen for all of human history and understood for only the last two centuries. The ancients recorded the great storms as omens: the Chinese astronomers logged falling stars with the same diligence they gave eclipses, the records reaching back more than two millennia and preserving, in court annals and dynasty histories, the nights when stars fell like rain; the European chronicles kept the same habit, the medieval monasteries noting the fiery signs with alarm and prophecy; and every culture read its own meaning into the streaks, the sky's fireworks being scripture's raw material long before they were science's. The prehistory's puzzle is the misunderstanding: for two thousand years the learned opinion held that meteors were atmospheric, weather rather than astronomy, the streaks being classed with lightning and the shooting star being literally a star only in the poets' license.

The modern discovery begins with the storm of eighteen thirty-three, the night the sky over North America broke into the most recorded meteor display in history. The Leonid storm of November twelfth descended on a waking nation, the meteors falling at rates later estimated in the tens of thousands per hour, the radiant in Leo riding the sky like a hub whose spokes were the streaks; the witnesses' letters and the newspapers' columns preserved the shock, and the Yale professor Denison Olmsted did the decisive science, collecting the reports like a detective and demonstrating that the display had a geometry: the meteors had radiated from a single point, the event had been of short duration, and it had not been seen from Europe, the conclusion being that the sky had passed through a cloud of particles in space, the shooting star being astronomy after all.

The comet connection came a generation later and completed the discovery. Giovanni Schiaparelli, working in Milan in eighteen sixty-six, computed the orbit of the Perseid meteor stream and found it matched comet Swift-Tuttle, the shower being the comet's shed dust encountered annually; the Leonids were matched the same year to comet Tempel-Tuttle, the great storm's parent being found; and the astronomers learned that the showers are the comets' wakes, the annual displays being the sky's most visible evidence that the Earth shares its neighborhood with dirty ice. The discovery chapter closes with the paradox the rest of the book explores: the phenomenon every human has seen, the one that needed no instrument, was the last major sky event to be understood, the atmosphere's own noise having masked the astronomy for two thousand years.

The nineteenth century's discovery story deserves its full arc, the century that moved the meteor from weather to

astronomy in three generations. The eighteen thirty-three storm opened the case, Olmsted's letters having shown the radiant and the cloud; the eighteen sixties closed it twice, the comet connections and the moon's own entry, the meteoritic science having been born the same decade when the falls of the stone showers persuaded the doubters; and the century's institutional tail being the visual networks' founding, the British and American associations having organized the counts that the showers' statistics required, the discovery having been converted from anecdote to data, the arc being the prehistory's answer, the two-thousand-year misunderstanding having been dismantled in thirty-three years, exactly one Leonid cycle, the sky having set the deadline and the science having met it.

The prehistory deserves its own closing page, the record before the record. The Chinese annals being the world's oldest meteor archive, the court astronomers having logged falling stars with dates and radiants for two millennia, the storm years of the Leonids being countable in the old pages; the Islamic and the Indian and the Japanese records being the same habit in other scripts, the night's events having been state business wherever the sky was watched officially; and the European gap being the prehistory's own puzzle, the monastic chronicles keeping the storms but the learned tradition's having filed the streaks under weather, the two-thousand-year archive being the field's basement and the misunderstanding being the academy's, the chapter closing on the irony the whole book inherits: the peasants knew the sky was falling, the professors knew it was not, and the professors were the only ones who wrote the textbooks.

The Biela comet's story earns a place in the discovery record, the stream being the proof of the parent's death. The comet having been discovered in seventeen seventy-two, recognized as periodic in eighteen twenty-six, and observed splitting in eighteen forty-six, the twin fragments having returned as a pair and then vanished entirely by the eighteen sixties; and the Andromedid shower's great storms, the displays of the eighteen seventies being the fragments' own funeral, the stream's having been identified with the lost comet's dust, the display's later century-long fading being the trail's own aging, the case being the discovery chapter's most instructive: the shower being the parent's gravestone and epitaph, the sky's having recorded a comet's death in falling stars, the record's lesson being the book's own theme, the stream outliving the body, the dust being the durable page.

The instrumentation of the discovery deserves the chapter's last page, the nineteenth century's having been the eye's century and the camera's apprenticeship. The great storm of eighteen thirty-three having been the unaided eye's supreme data night, the crowd's letters being the raw archive; the studio photographers' having caught the streaks by the eighteen eighties, the plate's being the first instrument that never blinked; and the visual networks' standardized forms being the era's own invention, the discovery's having been systematized before it was mechanized, the chapter's lesson being the sequence this series keeps finding: the science's best instrument having been the committee, the observers' coordinated patience having beaten the lone genius to every discovery in this chapter, the meteor's history being the teamwork's own monument.

The chapter's penultimate entry belongs to the denial itself, the skeptical case having been the discovery's own obstacle course. The eighteenth-century academies' having demanded witnesses of standing, the peasants' testimony having been weighed and refused, the stone falls' having been dismissed as lightning's residue and the showers as marsh lights; the Leipzig professor's having published the meteorite-fall reports as peasant lies, the skeptical literature's having been confident and wrong; and the reversal's having been complete within a generation, the eighteen-oh-three French falls having been examined by the commission, the stones' being declared genuine, the academy's having published its own recantation, the entry being the discovery chapter's own epistemology lesson, the evidence's having won by persistence rather than rank, the subject's history being the standing reminder that the well-credentialed no is not an argument.